

```
In [9]: from bs4 import BeautifulSoup
import requests
import pandas as pd

url = 'https://www.coingecko.com/'
response = requests.get(url, headers={"User-Agent": "Mozilla/5.0"})
soup = BeautifulSoup(response.text, 'html.parser')

table = soup.find('table')
headers = [th.get_text(strip=True) for th in table.find('tr').find_all('th')]

rows = []
for tr in table.find_all('tr')[1:]:
    cols = tr.find_all('td')
    if not cols:
        continue

    # indexes 4,5,6 correspond to Price Change 1h, 24h, 7d columns (based on site layout)
    percent_cols = [4, 5, 6, 7]

    row_data = []
    for idx, td in enumerate(cols):
        text_value = td.get_text(strip=True)
        if idx in percent_cols:
            i_tag = td.find('i')
            sign = ''
            if i_tag:
                classes = i_tag.get('class', [])
                if any('fa-caret-up' in c for c in classes):
                    sign = '+'
                elif any('fa-caret-down' in c for c in classes):
                    sign = '-'
            row_data.append(sign + text_value)
        else:
            row_data.append(text_value)

    rows.append(row_data)

df = pd.DataFrame(rows, columns=headers)
df.head()
```

Out[9]:

	#	Coin	Price	1h	24h	7d	30d	24h Volume	Market Cap	FDV	Market Cap / FDV	Last 7 Days
0	1	BitcoinBTC	Buy \$120,512	+0.4%	+1.5%	+4.9%	2.9%	\$54,346,483,203	\$2,398,650,621,343	\$2,398,650,621,943		1.0
1	2	EthereumETH	Buy \$4,346.32	+1.3%	+2.8%	+19.4%	49.2%	\$36,287,364,448	\$524,326,176,431	\$524,326,176,431		1.0
2	3	XRPXRP	Buy \$3.21	+0.8%	+0.4%	+5.7%	16.1%	\$6,648,510,551	\$190,241,185,840	\$320,720,791,470		0.59
3	4	TetherUSDT	Buy \$0.9999	+0.0%	+0.0%	+0.0%	0.0%	\$107,128,887,153	\$164,515,160,286	\$164,515,160,286		1.0
4	5	BNBBNB	Buy \$808.90	+0.9%	+0.9%	+6.4%	18.7%	\$1,622,326,073	\$112,572,455,702	\$112,572,455,702		1.0

```
In [10]: df=df.drop(columns=['#','#','Last 7 Days'], errors='ignore')
```

```
In [11]: for col in df.columns:
    if df[col].astype(str).str.contains(r'[\$%]').any():
        df[col] = (
            df[col].astype(str)
            .str.replace('$',' ', regex=True)
            .str.replace(r',', '', regex=True)
            .str.replace('%',' ', regex=True)
        )
    # Convert to float
    df[col] = pd.to_numeric(df[col], errors='coerce')
```

```
In [12]: df.head()
```

Out[12]:

	Coin	Price	1h	24h	7d	30d	24h Volume	Market Cap	FDV	Market Cap / FDV
0	BitcoinBTC	120512.0000	0.4	1.5	4.9	2.9	5.434648e+10	2398650621943	2398650621943	1.0
1	EthereumETH	4346.3200	1.3	2.8	19.4	49.2	3.628736e+10	524326176431	524326176431	1.0
2	XRPXRP	3.2100	0.8	0.4	5.7	16.1	6.648511e+09	190241185840	320720791470	0.59
3	TetherUSDT	0.9999	0.0	0.0	0.0	0.0	1.071289e+11	164515160286	164515160286	1.0
4	BNBBNB	808.9000	0.9	0.9	6.4	18.7	1.622326e+09	112572455702	112572455702	1.0

```
In [13]: df.info()

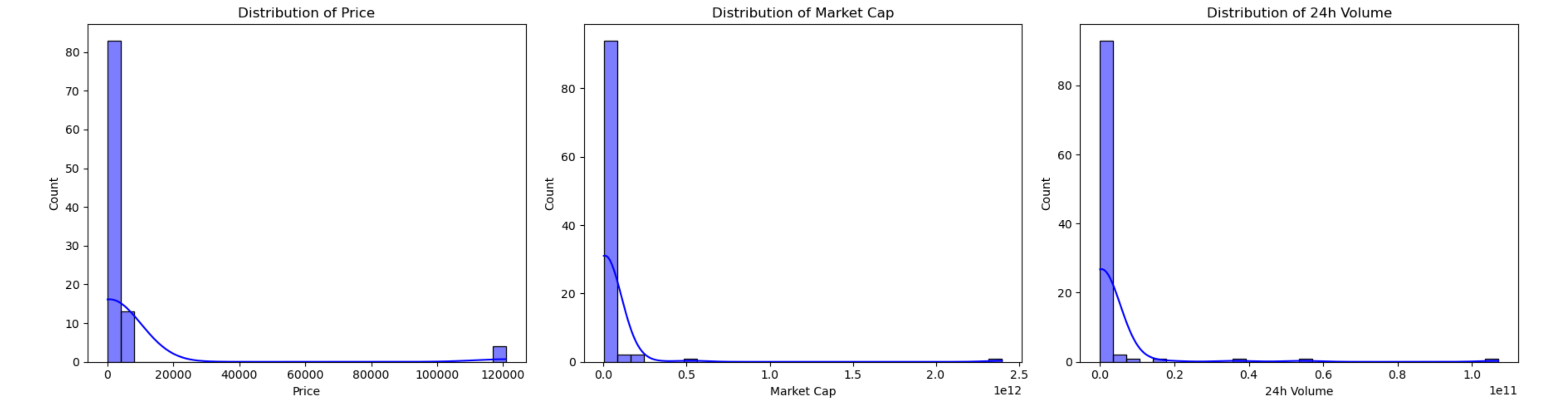
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 10 columns):
 #   Column      Non-Null Count  Dtype
---  --
 0   Coin        100 non-null    object
 1   Price       100 non-null    float64
 2   1h          100 non-null    float64
 3   24h         100 non-null    float64
 4   7d          100 non-null    float64
 5   30d         100 non-null    float64
 6   24h Volume  100 non-null    float64
 7   Market Cap  100 non-null    int64
 8   FDV         100 non-null    int64
 9   Market Cap / FDV  100 non-null    object
dtypes: float64(6), int64(2), object(2)
memory usage: 7.9+ KB
```

```
In [14]: df.describe()
```

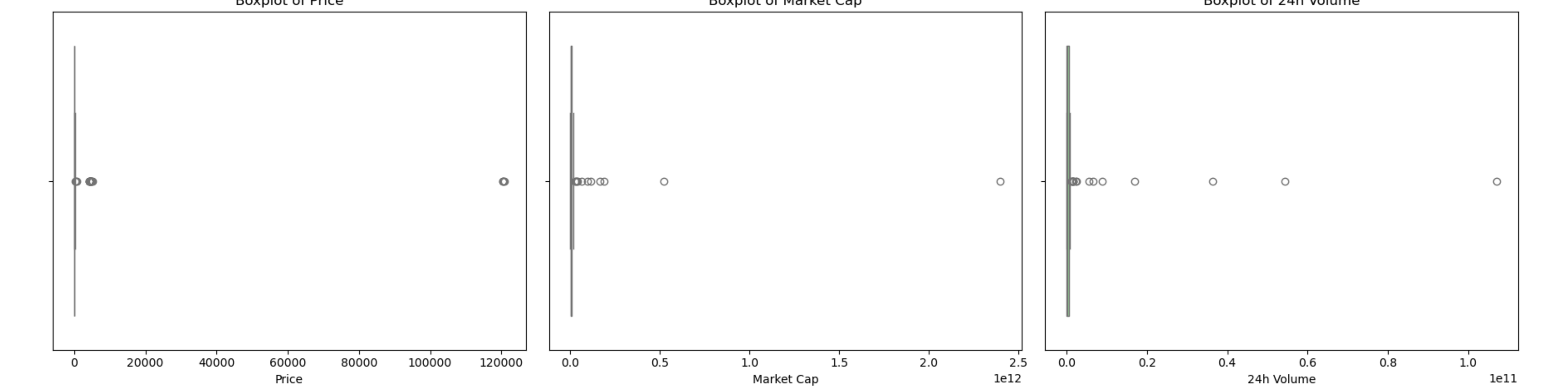
Out[14]:

	Price	1h	24h	7d	30d	24h Volume	Market Cap	FDV
count	100.000000	100.000000	100.000000	100.000000	100.000000	1.000000e+02	1.000000e+02	1.000000e+02
mean	5457.677079	0.909000	0.154000	8.589000	20.183000	2.652324e+09	4.088836e+10	4.606093e+10
std	23667.542986	0.864227	2.418256	9.765264	25.205162	1.251791e+10	2.453362e+11	2.466591e+11
min	0.000012	-1.700000	-6.700000	-11.800000	0.000000	0.000000e+00	1.316152e+09	1.316152e+09
25%	0.809900	0.375000	-1.025000	2.025000	2.275000	5.327854e+07	1.826430e+09	2.189593e+09
50%	3.300000	0.950000	0.000000	7.200000	13.100000	1.347671e+08	2.892591e+09	4.978945e+09
75%	139.037500	1.300000	1.275000	12.650000	29.875000	4.907604e+08	8.914795e+09	1.181755e+10
max	120788.000000	5.800000	14.600000	68.600000	158.700000	1.071289e+11	2.398651e+12	2.398651e+12

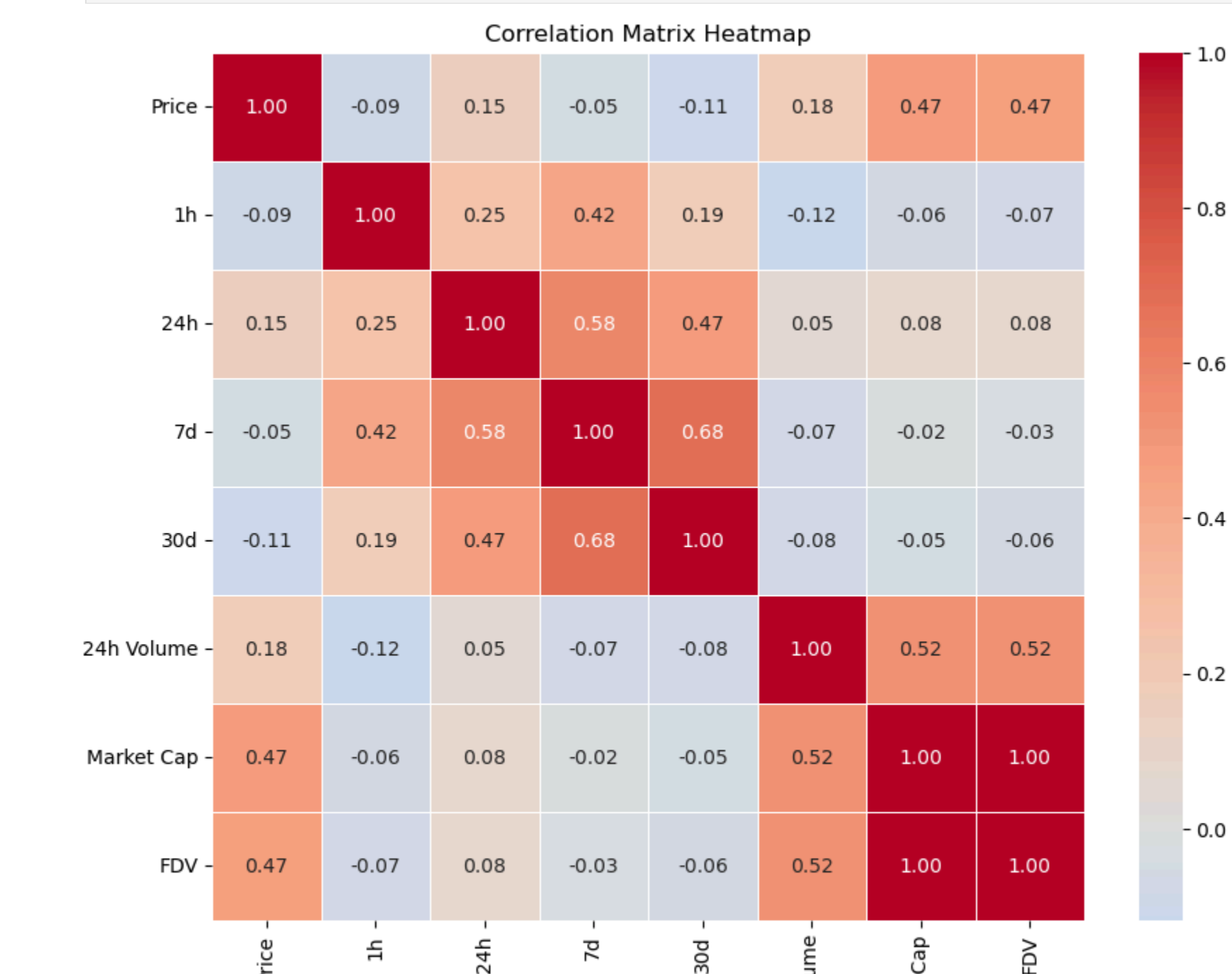
```
In [15]: import matplotlib.pyplot as plt
import seaborn as sns
numeric_cols = ['Price', 'Market Cap', '24h Volume']
plt.figure(figsize=(18, 5))
for i, col in enumerate(numeric_cols, 1):
    plt.subplot(1, 3, i)
    sns.histplot(df[col].dropna(), kde=True, bins=30, color='blue')
    plt.title(f'Distribution of {col}')
```



```
In [16]: plt.figure(figsize=(18, 5))
for i, col in enumerate(numeric_cols, 1):
    plt.subplot(1, 3, i)
    sns.boxplot(df[col].dropna(), color='lightgreen')
    plt.title(f'Boxplot of {col}')
```



```
In [17]: numeric_df = df.select_dtypes(include=['float64', 'int64'])
corr_matrix = numeric_df.corr()
plt.figure(figsize=(10, 8))
sns.heatmap(
    corr_matrix,
    annot=True,
    cmap='coolwarm',
    center=0,
    linewidth=0.5,
    fmt=".2f"
)
```



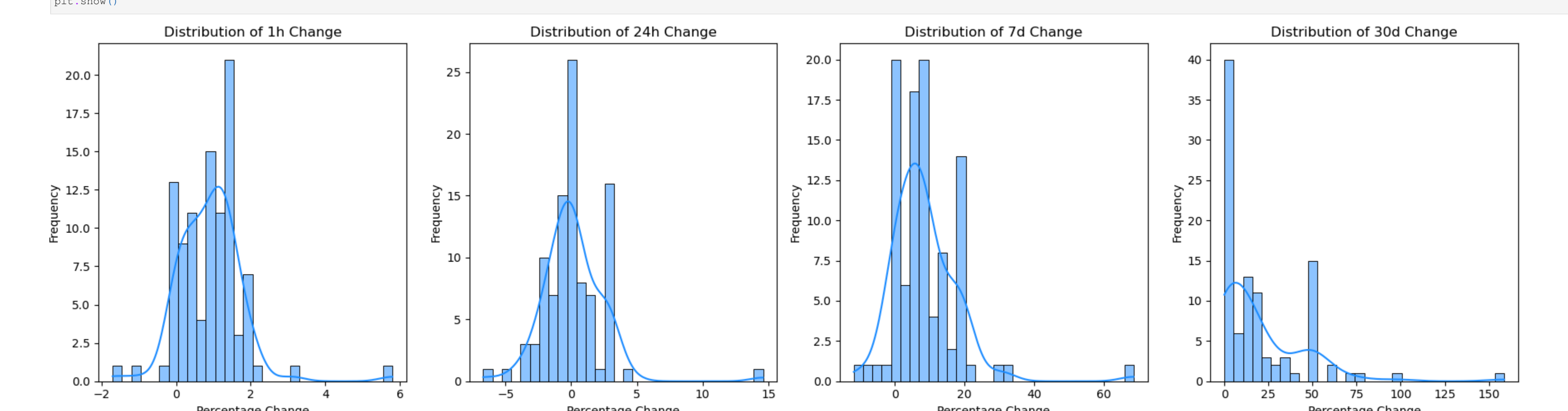
```
In [19]: strong_corrs = (
    corr_matrix.where(lambda x: (x.abs() > 0.4) & (x.abs() < 0.99))
    .stack()
    .reset_index()
    .rename(columns={0: 'correlation'})
    .sort_values(by='correlation', ascending=False)
)

print(f"Strong correlations (> 0.4 or < -0.4):")
strong_corrs
```

Out[19]:

	level_0	level_1	correlation
7	7d	30d	0.677863
9	30d	7d	0.677863
6	7d	24h	0.578957
3	24h	7d	0.578957
10	24h Volume	Market Cap	0.524856
13	Market Cap	24h Volume	0.524856
11	24h Volume	FDV	0.521205
15	FDV	24h Volume	0.521205
0	Price	Market Cap	0.473973
12	Market Cap	Price	0.473973
8	30d	24h	0.467794
4	24h	30d	0.467794
1	Price	FDV	0.466559
14	FDV	Price	0.466559
5	7d	1h	0.415631
2	1h	7d	0.415631

```
In [21]: percent_cols = ['1h', '24h', '7d', '30d']
plt.figure(figsize=(18, 5))
for i, col in enumerate(percent_cols, 1):
    plt.subplot(1, 4, i)
    sns.histplot(df[col].dropna(), kde=True, bins=30, color='dodgerblue')
    plt.title(f'Distribution of {col} Change')
    plt.xlabel(f'Percentage Change')
```



```
In [23]: for col in percent_cols:
    counts = {
        'positive': int((df[col] > 0).sum()),
        'negative': int((df[col] < 0).sum()),
        'zero': int((df[col] == 0).sum())
    }
    print(f"{col} change counts:")
    print(counts)

1h change counts:
{'positive': 84, 'negative': 4, 'zero': 12}
24h change counts:
{'positive': 45, 'negative': 45, 'zero': 12}
7d change counts:
{'positive': 85, 'negative': 7, 'zero': 8}
30d change counts:
{'positive': 93, 'negative': 0, 'zero': 7}
```

```
In [25]: print("\nTop cryptocurrencies by Market Cap:")
print(df[['Coin', 'Market Cap']].sort_values(by='Market Cap', ascending=False).head(10))

Top cryptocurrencies by Market Cap:
   Coin  Market Cap
0  BitcoinBTC  2398650621943
1  EthereumETH  524326176431
2  XRPXRP  190241185840
3  TetherUSDT  164515160286
4  BNBBNB  112572455702
5  SolanaSOL  9740335535
6  USDCUSDC  6504132883
7  Lido Staked EtherLSD  38439458782
8  DogecoinDOGE  34971593424
9  TRONTRX  32667049338
```

```
In [27]: print("\nTop cryptocurrencies by Price:")
print(df[['Coin', 'Price']].sort_values(by='Price', ascending=False).head(10))

Top cryptocurrencies by Price:
   Coin  Price
79  Lombard Staked BTCLBTC  120788.00
32  Coinbase Wrapped BTCBTCBTC  120532.00
0  BitcoinBTC  120512.00
12  Wrapped BitcoinWBTC  120476.00
31  Wrapped sETHINGSTETH  5259.31
63  Rocket Pool ETHETH  4938.54
16  Wrapped Beacon ETHWBETH  4682.46
84  Liquid Staked ETHLBETH  4659.82
18  Wrapped eETHETHETH  4658.77
65  Mantle Staked ETHETH  4628.95
```

```
In [29]: print("\nTop cryptocurrencies by Volume of last 24 h:")
print(df[['Coin', '24h Volume']].sort_values(by='24h Volume', ascending=False).head(10))

Top cryptocurrencies by Volume of last 24 h:
   Coin  24h Volume
3  TetherUSDT  1.071289e+11
0  BitcoinBTC  5.434648e+10
1  EthereumETH  3.628736e+10
6  USDCUSDC  1.693408e+10
```

```
99          First Digital USDFDUSD      8.824327e+09
2          XRPXRP      6.848511e+09
9          SolanaSOL      5.555509e+09
8          DogecoinDOGE      2.339007e+09
30  Binance Bridged USDT (BNB Smart Chain)BSC-USDT      2.271819e+09
4          BNBBSB      1.622526e+09

In [30]: df.to_csv('cryptocurrency_data.csv', index=False)
```

Key Insights

- 1. Bitcoin Dominance: In the market cap ranking plot, Bitcoin stands far ahead of other coins, occupying the top spot with a market cap that's significantly larger than the rest — the bar for BTC clearly dwarfs Ethereum's and other altcoins' bars.
- 2. The trading volume distribution plot shows that a few large-cap coins (like BTC and ETH and USDT) drive most of the total traded volume, while the vast majority of coins have relatively small daily volumes.
- 3. Performance Spread: The 24h, 7d and 30d percentage change histograms reveal a mixed market for today but a overall positive i.e. increase — with many coins clustered close to 0% change but a small number of outliers both on the positive and negative side, indicating pockets of high volatility for today but for longer range there's a clear dominance on the positive side showing a net increase in crypto market.